



Why is a 3-year NRM following allogeneic transplantation still stuck at approximately 20%?

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ABSTRACT

Whether and when to recommend an allogeneic stem cell transplant (SCT) for a patient with leukemia is a treatment decision that rests on determining whether the transplant or non-transplant option carries the greatest probability of 3–5-year survival. While SCT confers a greater possibility of leukemia cure, the decision to transplant has to be made in the light of the high chance of treatment-related mortality (TRM) that follows the allograft. Here we identify that current estimates of a 20% 3-year TRM hold largely true for a variety of leukemias, diverse types of conditioning regimen, and varied donor-recipient compatibility across a wide age-range. While there is a wide spectrum of causes of death in the first few months after SCT, they usually stem from a limited set of immediate post-transplant complications, including those induced by the conditioning regimen, post-transplant endovascular damage, gut dysbiosis, graft-versus-host disease, and immunodeficiency causing viral reactivation. As we better understand and improve preventative treatments for these initiating events there is a real expectation that TRM will continue to fall to levels well below 10% within the next decade.

1. Introduction

Allogeneic stem cell transplantation (SCT) as a treatment for acute leukemia has stood the test of time for nearly 50 years. In that period both transplant and nontransplant outcomes for leukemia have improved significantly. While new treatments superior to allogeneic SCT have emerged that have altered the selection of patients for transplantation, advances in our ability to control post-transplant complications have also led to improved outcomes and wider applicability. Today SCT achieve comparable outcomes in leukemia patients, whether the recipient is matched or mismatched with the donor, and irrespective of age, with successful SCT being routinely performed in the 7th - 8th decade. The decision for transplant versus nontransplant treatment has always depended on selecting the approach (based on current data) with the best survival for an individual at the moment in their treatment when a treatment choice has to be made (usually in first or second complete remission). SCT always tends to have a superior ability to cure the leukemia, but the advantage is offset by a greater treatment related mortality (TRM) compared with non-transplant approaches. Thus attempts to reduce TRM are critical for making SCT the treatment of choice for more patients with otherwise incurable leukemia. Major reductions in TRM would particularly alter decision making for high risk leukemias where new therapies have had less impact and where SCT has been shown to be curative.

Current decision-making for SCT rotates around a perceived TRM of around 20%. This review explores the validity of the 20% TRM benchmark across a variety of patient groups and transplant approaches, outlines the main drivers of TRM and current progress in developing new treatments to eliminate the provocative post-transplant events that precede non-relapse death after transplant.

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Table 1
Transplant-related mortality in 5 large multicenter retrospective analyses.

Study Group		Study era	Study description	N (total)	3 y TRM (95% CI)
EBMT	Savani et al. [2]	2000–2012	Peripheral blood SCT (PBSCT)	837	18 (16–21)
			vs bone marrowtransplant (BMT)	9011 (9848)	20 (18–23)
	Savani et al. [3]	2000–2012	Reduced intensity conditioning (RIC)	883	27.5 (24–30)
			vs myeloablative conditioning (MAC) (includes 926 patients older than 50y)	1041 (1924)	27.5 (25–30)
CIBMTR	Teira et al. [5]	2003–2010	Busulfan/Cytosan (BuCy)	514	22.9 (17.6–28.5)
			vs Cytosan/total body irradiation (TBI)	338 (852)	16.5 (12.0–21.6)
			CMV serostatus: donor (D)/recipient(R)		
			D + /R +	3253	17-28*
			D + /R-	1074	15-23*
			D-/R +	2607	15-23*
			D-/R-	2535 (9469)	14-23*

* Range: *Lowest* - chronic myelogenous leukemia; *intermediate* - acute leukemia; *highest* - myelodysplastic syndrome.

2. Post-transplant TRM in the 21st century

In the 1970s when large numbers of SCT began to be performed, TRM could reach 40%. Death from all causes of infection, including viral reactivation, graft-versus-host disease (GVHD) and conditioning regimen-related organ damage were the major contributors. Major improvements in TRM were achieved by the introduction of improved antivirals and antifungals and in particular the preemptive treatment of cytomegalovirus (CMV) to prevent pneumonitis, patient-adapted reduced intensity conditioning regimens, and better prevention of GVHD. These changes have been associated with a significant fall in TRM, such that by the beginning of this century the TRM reached around 20% [1]. This figure appears to hold in current reports from large multicenter databases that can be assumed to be an unbiased representation of the state of the art (with the proviso that the results of 3- or 5-year survival must be derived from cohorts of patients transplanted at least before 2015). Table 1 lists 4 such analyses from Europe (European Bone Marrow Transplantation Group – EBMT) and the predominantly USA-based data from the CIBMTR [2–5]. With the exception of CML, now seldom transplanted, TRM is comparable across different hematologic malignancies whether acute leukemia or MDS. What is remarkable is that TRM is largely comparable whether the transplant is matched or mismatched, whether the source of stem cells is derived from marrow, blood, or umbilical cord, or whether patients are adults or older adults (over 60 years). Children have been excluded from this analysis but have significantly lower TRM. As shown by Teira et al. [5], prior exposure to CMV is a consistent

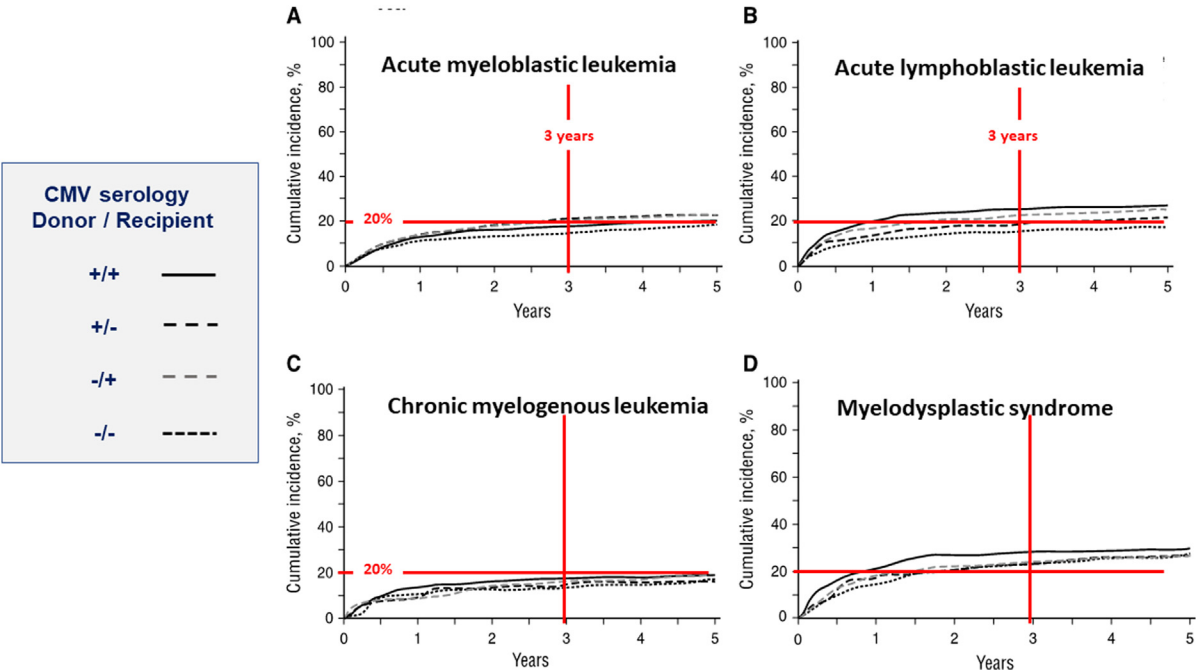


Fig. 1. Comparison of NRM between different disease types and CMV serologies showing 3- year 20% TRM markers [5] Republished with permission of American Society of Hematology, from Early cytomegalovirus reactivation remains associated with increased transplant-related mortality in the current era: a CIBMTR analysis, Teira P, Battiwalla M, Ramanathan M et al. Volume 127, No. 20, 2016; permission conveyed through Copyright Clearance Center, Inc.

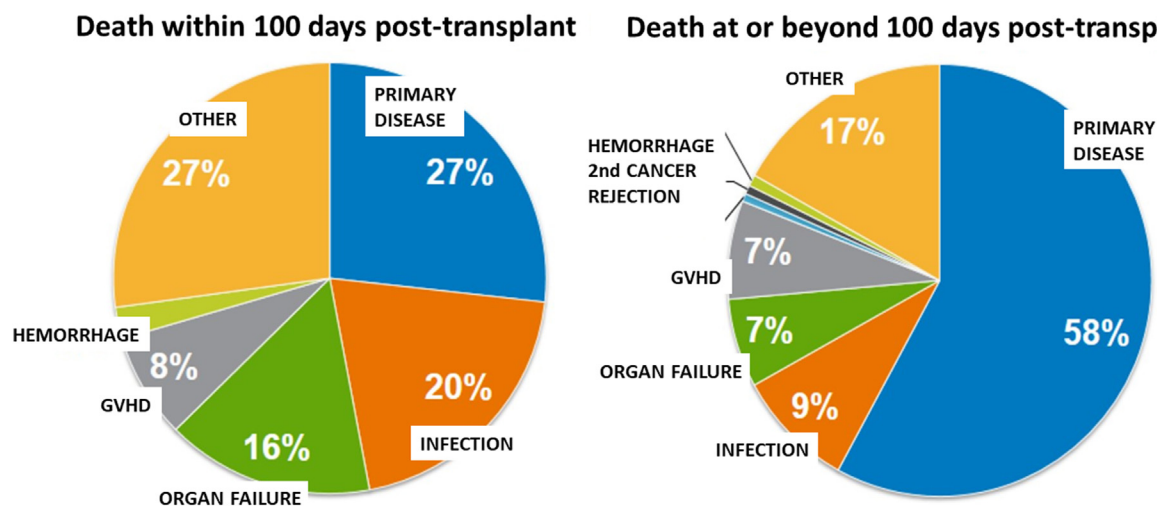


Fig. 2. Causes of transplant related mortality (CIBMTR [8]).

negative factor for survival but nevertheless there is only a 7% difference in TRM between the worst group (donor and patient CMV positive) and the most favorable group (donor and patient CMV naive) (Fig. 1). This data confirms that a value of 20% TRM is indeed a reliable, widely applicable benchmark for basing treatment decision. That said, it should be noted that individual transplant centers frequently achieve TRM lower than those reported in multicenter databases. Sohl et al. for example report a 5-year TRM of 15% in a series of patients transplanted at a single center between 2005 and 2015 [6]. Also, there is growing evidence indicating that it is possible to achieve 1-year TRM around 5% even when a haploidentical relative is used as the donor [7]. Thus it is realistic to anticipate much lower TRM as advances in post-transplant care spread into general transplant practice.

3. The post-transplant complications that drive TRM

Fig. 2, reproduced from the CIBMTR Summary Slides 2017, identifies firstly that the main cause of treatment failure after SCT is disease relapse [8]. The causes of treatment failure in the first few months after transplant are predominantly TRM from GVHD, organ failure and infection, as well as from a large category of causes termed “other.” It should be appreciated, however, that such pie diagrams fail to describe the pathways to multi-organ failure and death. Thus, while the immediate cause of death may fall into the “other” category, a restricted number of provocative causes were primarily responsible. These initial complications are centered around engraftment, immune dysfunction, infection, GVHD and conditioning regimen toxicity. They set in motion a series of events, which include adverse consequences of treatment used to manage them, which lead to an inability to navigate therapeutically between efficacy and toxicity and subsequently to death of the patient. Fig. 3 models this interplay between first order complications, the chain reaction of subsequent complications, some induced by therapy, leading to treatment failure, a pattern woefully familiar to stem cell transplant physicians.

4. Progress towards preventing TRM

The implication of this tendency after SCT for initial complications to lead to a downward spiral of failure to control events is that TRM can be best addressed by treating or better preventing outright the well-known post-transplant complications of immune dysregulation, organ damage, GVHD, infection and viral reactivation. Summarized here are the recent developments in understanding transplant-related pathophysiology and new lines of treatment that have the potential to eliminate the common life-threatening complications that follow SCT (Table 2).

Gut flora dysbiosis after SCT. It is now clear that alterations in the gastrointestinal flora play a significant role in the early adverse events that follow SCT. Mucosal damage and reduced oral nutrition induced by the conditioning regimen is a provocative factor for changes in the composition of the gut microbiota. Conditioning-regimen-induced neutropenia and antibiotics used to treat neutropenic infection promote bacterial invasion into the tissues as well as profoundly disturbing the natural ecology of the many species of bacteria residing in the gut. These changes result in “dysbiosis” of the gut microflora, characterized by a loss of bacterial species diversity and of “good” bacteria—in particular *Blautia* anaerobes—and an increase in proteobacteria, *streptococci*, *enterococci*, and *clostridium difficile*. These alterations in the gut microflora set the stage for inflammatory processes that further damage the gut mucosa and provoke GVHD and delayed immune reconstitution. It is now recognized that gut dysbiosis is a significant contributor to increased post-transplant mortality. Understanding dysbiosis has led to strategies now under development to counteract the damage through selective use of antibiotics, “transplants” of favorable bacteria, and modification of bacterial metabolism (reviewed in Shono et al. [9]).

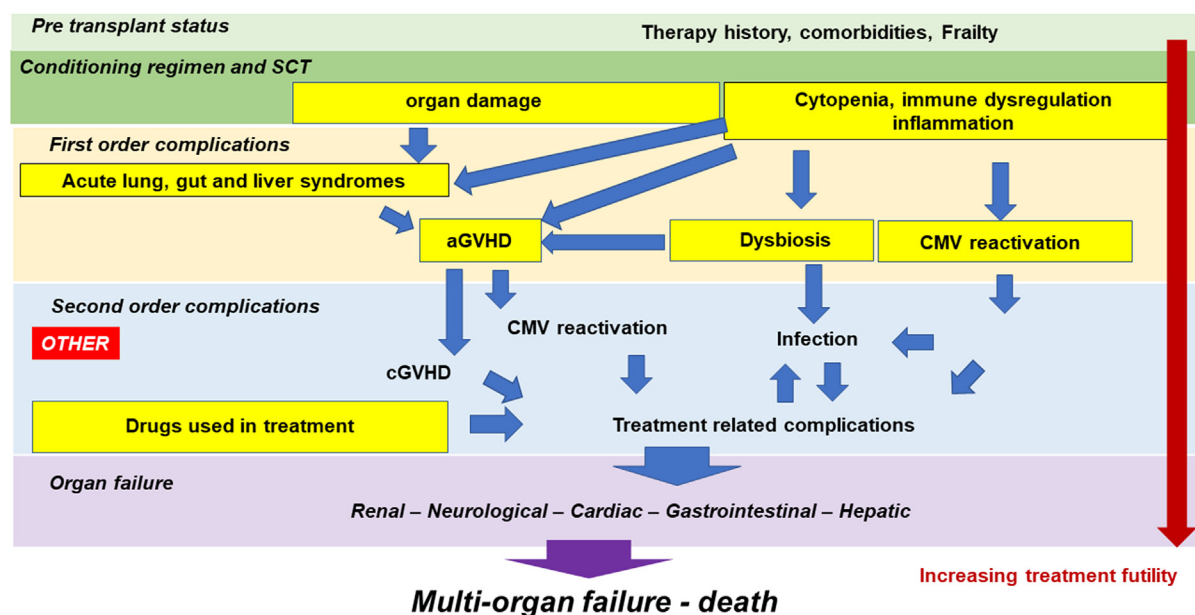


Fig. 3. The chain reaction of events that cause TRM.

Table 2

Applications of biomonitoring for predicting and managing post-transplant complications.

Timepoint	Endothelial damage	Damage repair	Dysbiosis	Viruses CMV, Ad, EBV, BK/JC polyomaviruses	Innate immune activation	GVHD
Pretransplant	Angiopoietin-2 VEGF- α PIGF Follistatin	Follistatin Endothelial growth factor Citruiline	Flora profile picornaviruses	PCR/antibodies	TNF α , IL-6 IL-1 β , sDNAM	Donor T _{regs} IL-33
Post transplant weekly	ST2 REG3 α Angiopoietin-2/ VEGF ratio Follistatin	REG3 α , HGF Micro RNAs Fecal calprotectin α -1 antitrypsin Endothelial growth factor Follistatin Amphiregulin	Blautia species Urinary 3-IS Viruses	PCR	TNF α IL-6 CRMf44	T _{regs} CD4/CD8 ratio IL-2R α TNFR-1 IL-8, IL-7 ST2 Sol TIM3 TNFR-1 Reg3 α

Key: ST2, CMV – cytomegalovirus; Ad - adenovirus; EBV - Epstein-Barr virus; T_{regs}-regulatory T cells; see ¹⁶ for full description of other abbreviations Adapted from He and Holtan¹⁶.

GVHD. In the absence of novel and effective immunosuppressive drugs, steroid treatment remains the mainstay of acute GVHD treatment. About a third of patients developing acute GVHD become steroid refractory and a third of these will die from complications of GVHD or its treatment. Preventing GVHD rather than treating it appears to be the best way forward. Graft manipulation can prevent GVHD. Techniques already in clinical practice or under development include T-cell depletion, selective T-cell depletion, generating suicide gene-modified T cells (switched off at the onset of GVHD), and infusions of regulatory T cells. Alternatively, GVHD can be blocked by preventing homing of grafted T cells to target organs, blocking antigen presentation to the donor T cells, or by preventing activation of allostimulated T cells. Lastly GVHD-related tissue damage induced by cytokines induced by activated T cells can be prevented by blocking IL6 with tocilizumab (reviewed in Kekre et al. [10] and Servais et al. [11]) (Fig. 4).

CMV reactivation. Although there is some data that CMV seropositivity can confer lower rates of leukemic relapse in selected patients, the overall impact of CMV reactivation on transplant outcome is negative. In the largest study to date, involving over 6000 SCT recipients, Teira showed that the occurrence of past CMV infection in recipient or in both donor and recipient was detrimental to post-transplant survival. CMV exposure accounted for a 5%–10% increase in TRM across all transplant disease types (acute leukemias, chronic myelogenous leukemia and myelodysplastic syndromes) [5]. In a smaller study, we found that the event of CMV reactivation post SCT was itself deleterious, resulting in more complications and longer hospitalizations than in transplant recipients who were CMV seropositive but did not reactivate CMV [12]. Together these studies point towards the importance of preventing CMV reactivation post graft so as to avoid downstream complications related to antiviral therapy. New, more effective and less toxic antiviral drugs, such as letermovir, and increasing application of virus-specific T cells and exploration of their preventative use, is an important step forward in reducing mortality from this virus as well as from other common viruses that reactivate after SCT [13].

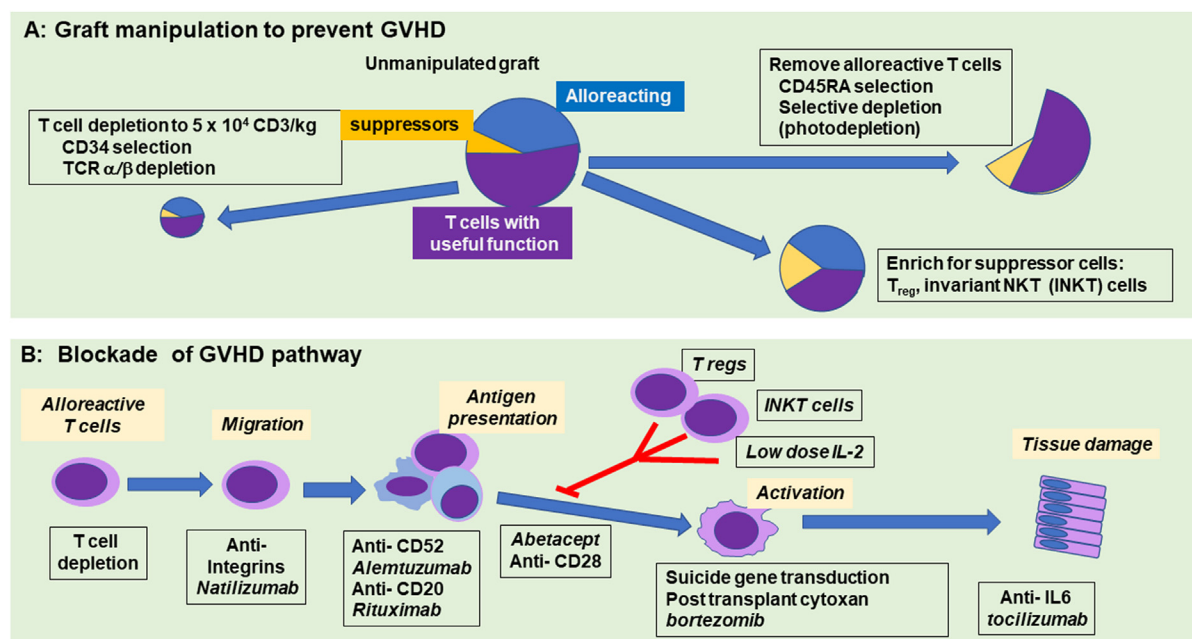


Fig. 4. Prevention of GVHD

Organ damage. While contributing to organ damage, conditioning regimen-related tissue damage has been much reduced by judicious reductions of conditioning regimen intensity according to the recipients' perceived frailty and co-morbidities. Better dosing of busulfan and strategies to deliver irradiation focused on the marrow and avoiding the lungfields have greatly reduced organ toxicity from conditioning regimens [14]. Currently, while conditioning regimens may contribute towards later organ failure, it is post-transplant organ damage that is harder to manage because the causes are multifactorial and the etiology is poorly understood. Thrombotic microangiopathy (TMA) veno-occlusive disease (VOD), and pulmonary failure are all potentially fatal problems. It is now emerging that endovascular damage is the provocative event leading to these syndromes. Initial studies using defibrotide to counteract vascular damage are promising [15].

5. Prospects for further reductions in TRM

Transplant outcomes never remain static. Much lower TRM are increasingly reported by individual transplant centers. Together with better prevention of complications with the emerging strategies described above, it is realistic to anticipate a global fall in TRM to between 5% and 10% in the next 5 years. Biomarker profiling early post-transplant will permit patient-specific fine tuning of preventative treatments to further limit complications before they arise and cause life-threatening complications. Better prediction could reduce TRM through tailored preventative management [16].

6. Conclusion

Accurately weighing the best treatment option for the individual patient with leukemia requires an awareness of contemporary data and an appreciation that the goalposts for treatment success are constantly shifting, both for non-transplant and transplant approaches. In future the minimization of treatment-related mortalities from all treatment approaches will focus management decisions for acute leukemia on the curative efficacy of each treatment. For its curative potential, allogeneic SCT still has the edge over non-transplant treatments in many situations. Nevertheless, much still remains to be achieved before the graft-versus-leukemia effect, intrinsic to the curative potential of the allograft, can be augmented sufficiently to prevent all post-transplant disease relapse.

Disclosures

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